

Supplementary Materials for
*A Multi-Fidelity Tensor Emulator for Spatiotemporal Outputs:
 Emulation of Arctic Sea-Ice Dynamics*

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Supplementary Figures

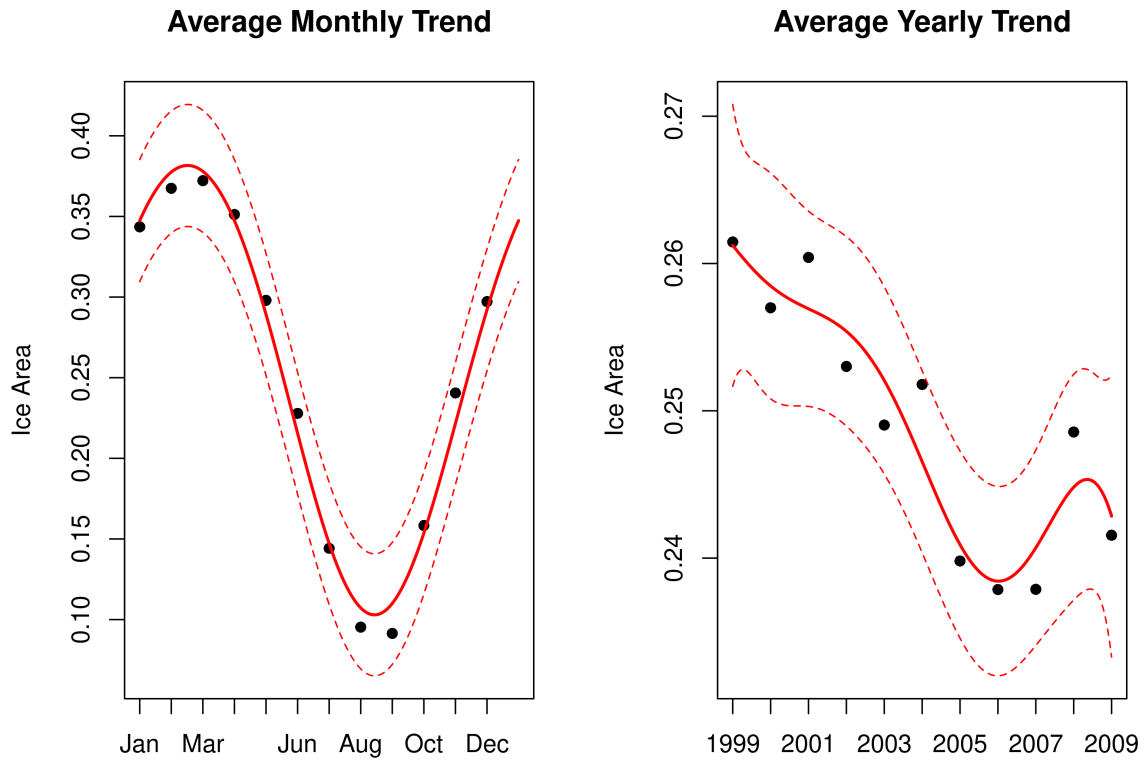


Figure S1: Temporal averages of MPAS-Seaice output, computed monthly and yearly, averaged over all spatial locations. Gaussian processes were fit to each trend: a periodic kernel for the monthly trend and an exponential kernel for the yearly trend. The fitted trends are shown in red, with dotted lines representing 95% credible intervals. Referenced in Section 2 of the main text.

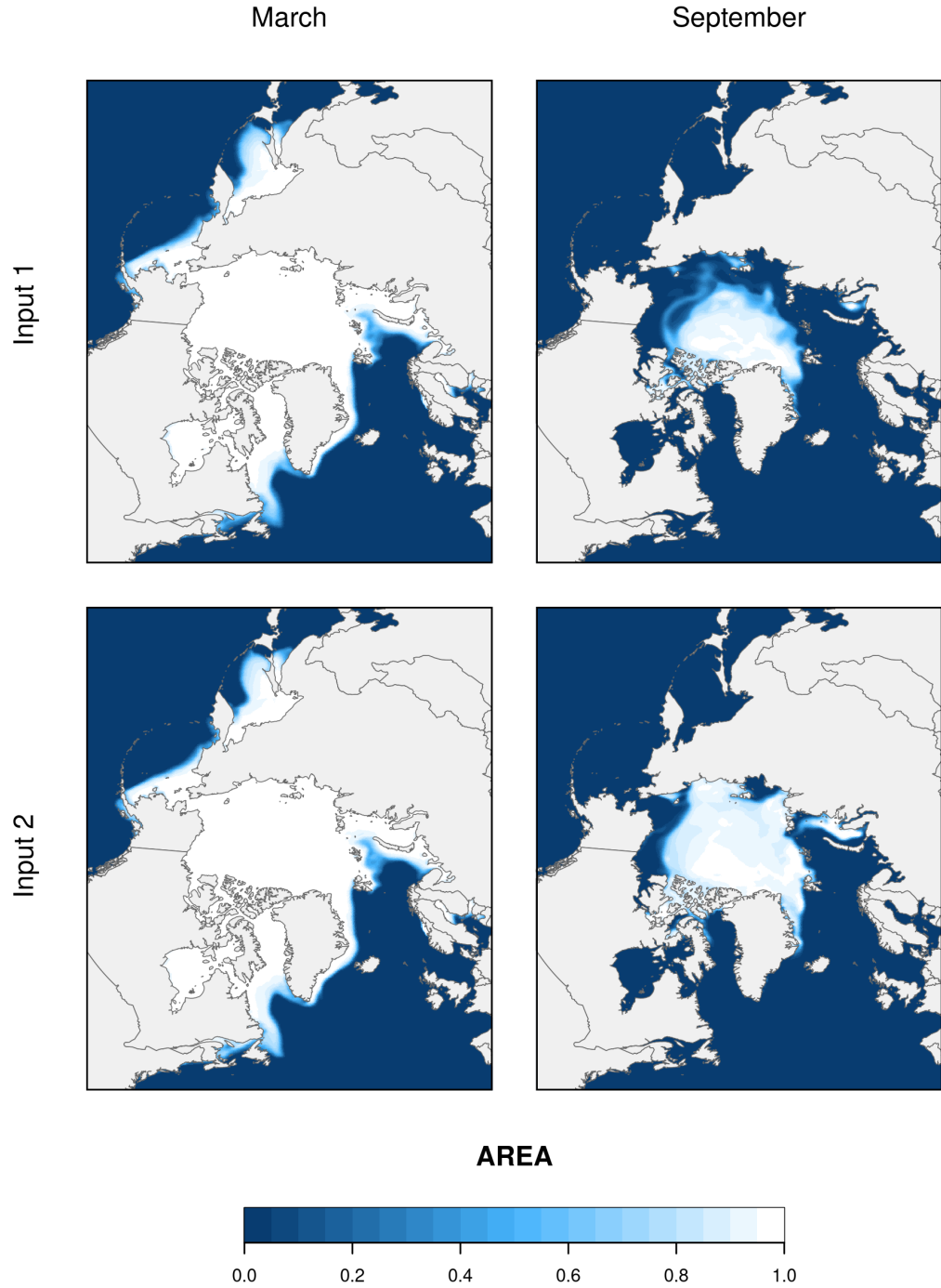


Figure S2: Example sea-ice concentration output for two MPAS-Seaice runs, March and September at high resolution. Colder months (e.g., March) are relatively consistent in spatial outputs, with small variability in transitional bands between ice and no ice. Warmer months (e.g., September) exhibit greater variability in the Central Arctic and surrounding seas. Referenced in Section 2 of the main text.

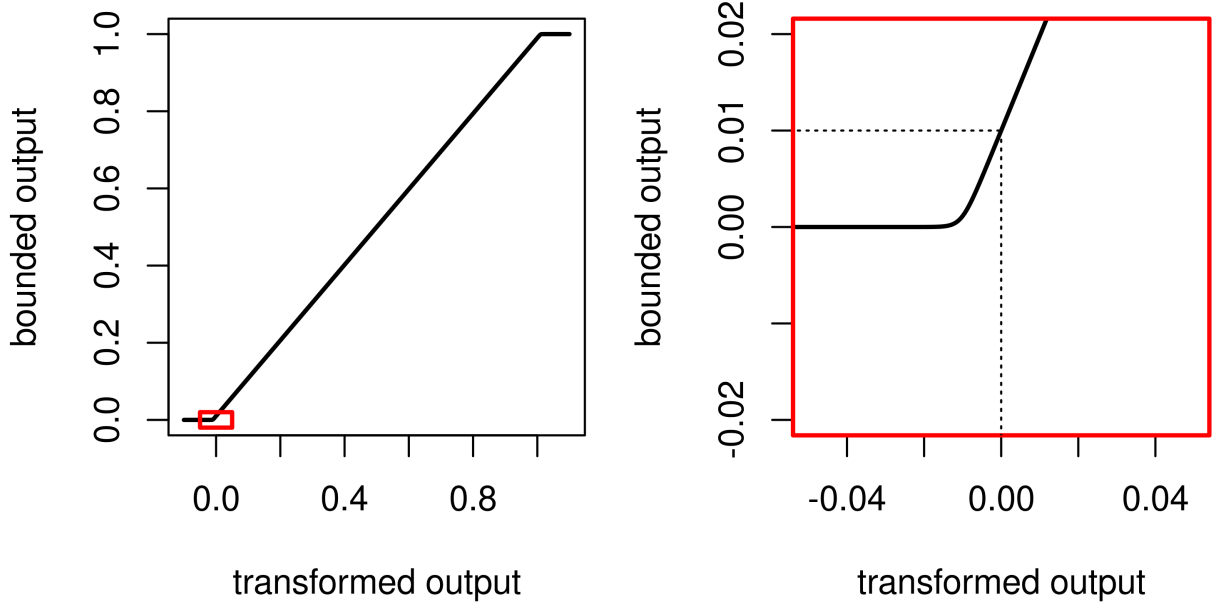


Figure S3: Logit-based transformation of $[0, 1]$ -bounded MPAS-Seaice outputs to the real line. Values are truncated to $[0.01, 0.99]$ and then mapped using a compressed logit function, resulting in a near-linear mapping between 0.01 and 0.99. The left panel shows the full transformation, and the right panel a zoomed-in view. This ensures that the data are suitable for the proposed methodology, which is broadly applicable to real-valued data and enables the use of basis decomposition methods such as Tucker decomposition. Referenced in Section 2 of the main text.

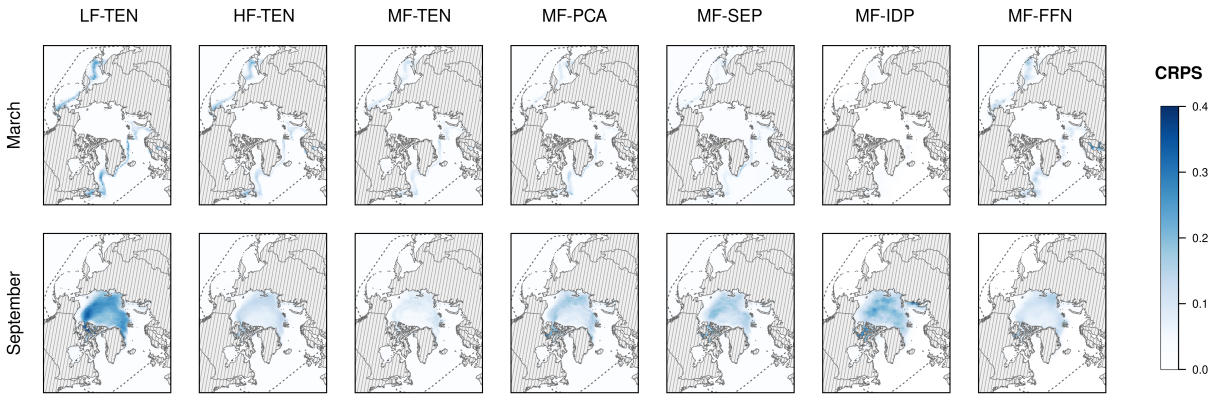


Figure S4: Spatial performance comparison of emulators on MPAS-Seaice data, showing continuous ranked probability score (CRPS) at each spatial location, averaged over leave-one-out cross-validation folds and years and shown separately for March and September. The MPAS-Seaice domain is outlined with a dotted line. Referenced in Section 5 of the main text.

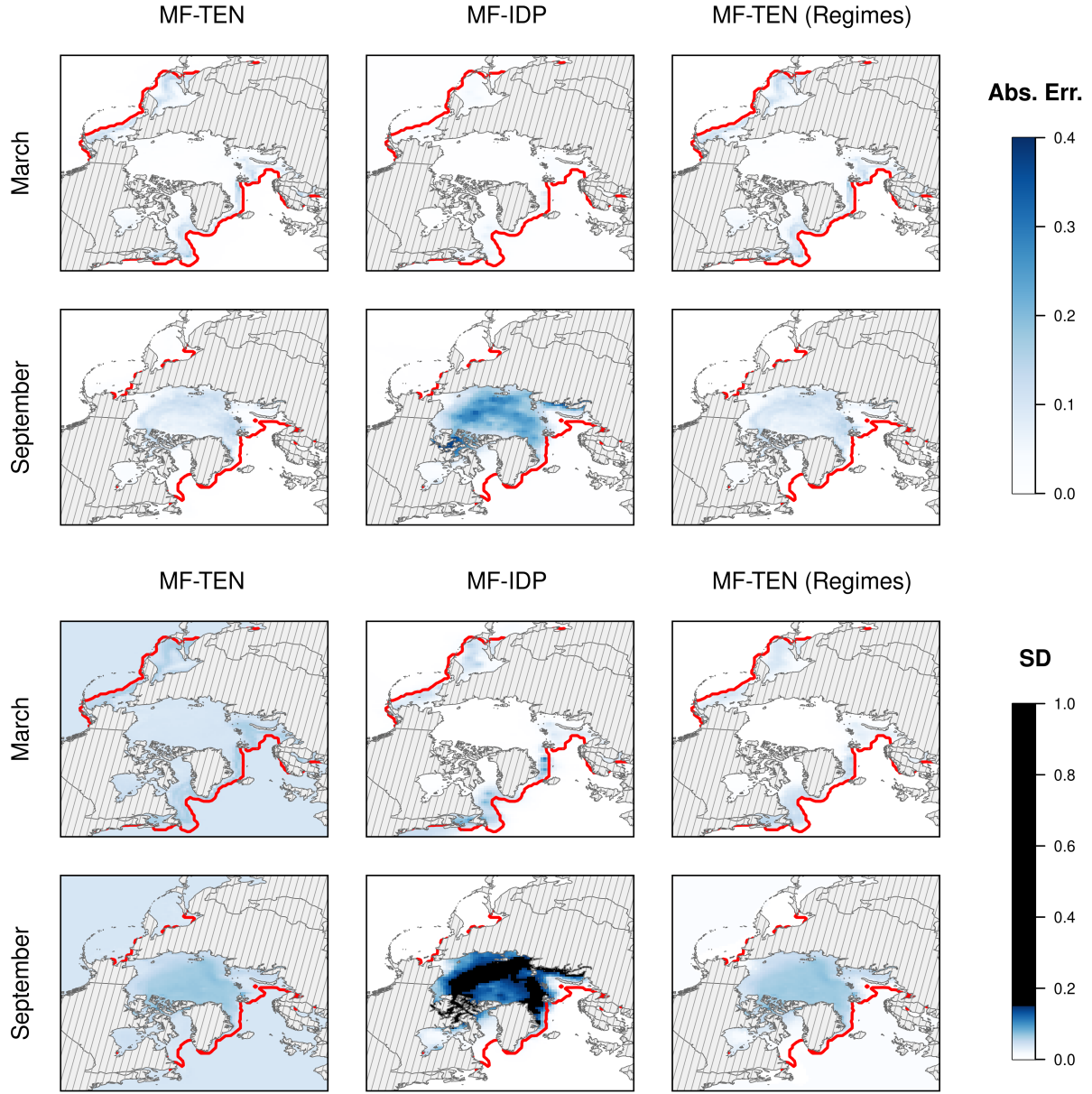


Figure S5: Spatial patterns of absolute error and standard deviation (SD) comparing the regime-based emulator with MF-TEN and MF-IDP. We consider four regimes: two spatial regions for warm months and two for cold months. The red line indicates the spatial regime boundary, defined separately for cold and warm months. The dotted line indicates the overall boundary of MPAS-Seaice. The regime-based emulator matches MF-TEN performance in spatially variable regions and MF-IDP performance in spatially constant regions. Referenced in Section 5 of the main text.

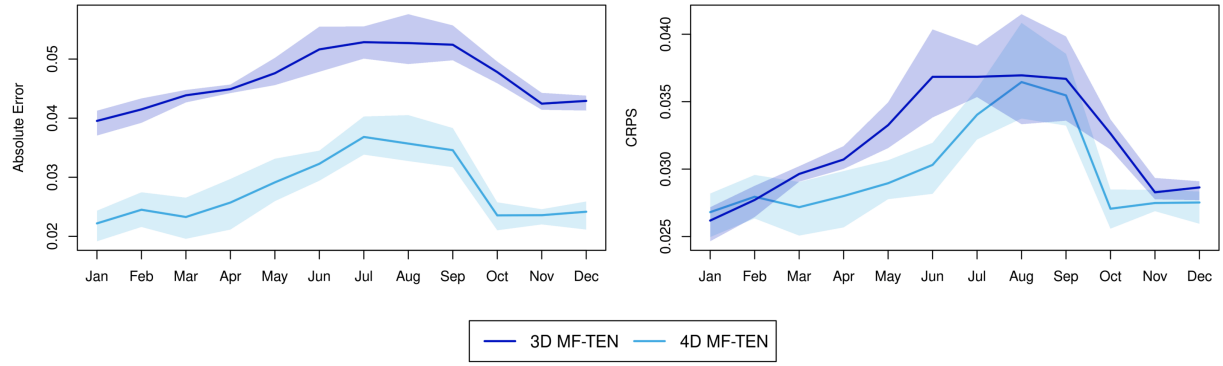


Figure S6: MF-TEN emulators for MPAS-Seaice data, comparing applications to third-order (spatial location, time, design space) and fourth-order (spatial location, month, year, design space) tensors. Plots show monthly absolute error and continuous ranked probability score (CRPS), averaged over spatial locations and years within each LOO-CV fold, with bands giving the 2.5%–97.5% quantile range across folds. Referenced in Section 6 of the main text.

A MPAS-Seaice Model Details

MPAS-Seaice [Turner et al., 2022] is the sea-ice component of the Energy Exascale Earth System Model (E3SM) [Leung et al., 2020]. The model is designed to simulate sea-ice evolution on unstructured meshes and is routinely used within fully coupled and standalone configurations.

The model includes a comprehensive representation of sea-ice physics, including vertical thermodynamics based on mushy layer physics [Turner and Hunke, 2015], shortwave radiation penetration and absorption [Briegleb and Light, 2007], sea-ice biogeochemistry [Jeffery et al., 2020], tracer transport [Lipscomb and Hunke, 2004], melt ponds [Hunke et al., 2013], ridging and mechanical redistribution [Lipscomb et al., 2007], and solution of the sea-ice momentum equation [Turner et al., 2022].

We conducted simulations using E3SM version 3.0.2 [E3SM Project, 2024] in standalone mode for the Northern Hemisphere and forced the model with six-hourly atmospheric fields for air temperature, air specific humidity, and 10 m air velocity from the JRA-3Q reanalysis [Kosaka et al., 2024], along with monthly climatologies of cloudiness [Röske, 2001] and precipitation [Griffies et al., 2009]. Downwelling shortwave radiation was calculated from cloudiness using the Arctic Ocean Model Intercomparison Project shortwave forcing formula [Hunke et al., 2015], while downwelling longwave radiation was calculated following Rosati and Miyakoda [1988]. Oceanic forcing inputs of sea surface salinity, initial sea surface temperature, currents, deep ocean heat flux, and sea-surface gradient were obtained from a monthly mean of 20 years of a Community Climate System Model climate run [Collins et al., 2006].

B Tucker Decomposition Example

In this section, we present a visual example of Tucker decomposition. For simplicity, we consider a three-dimensional tensor (space, time, design) rather than the four-dimensional tensors used elsewhere in the manuscript. Specifically, we model data with $n_s = 20$ spatial locations, $n_t = 10$ time points, and $n_x = 6$ simulations, resulting in a tensor $\mathcal{Z} \in \mathbb{R}^{20 \times 10 \times 6}$. Figure S7 shows spatiotemporal slices of this tensor. We will decompose $\mathcal{Z}$ via Tucker decomposition with ranks $(r_s, r_t, r_x) = (2, 2, 2)$ in this example.

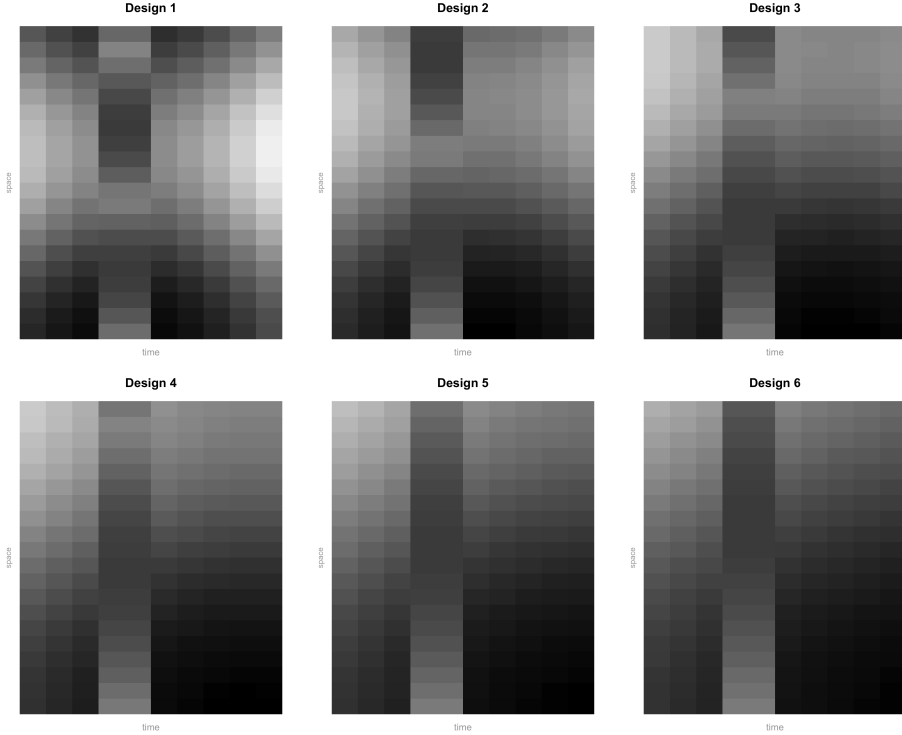


Figure S7: Slices of the example tensor $\mathcal{Z}$ along the third mode corresponding to design space. For $t = 4, 5$, the outputs have a different spatial pattern.

B.1 Higher-Order Orthogonal Iteration

We perform the Tucker decomposition using Higher-Order Orthogonal Iteration (HOOI), which iteratively minimizes the reconstruction error by updating the factor matrices. This yields orthonormal factor matrices that jointly minimize the tensor reconstruction error across all modes, converging to a local minimum of the approximation error [De Lathauwer et al., 2000]. Specifically, for a tensor $\mathcal{Z} \in \mathbb{R}^{n_s \times n_t \times n_x}$, we seek a core tensor $\mathcal{G} \in \mathbb{R}^{r_s \times r_t \times r_x}$ and factor matrices $U_s \in \mathbb{R}^{n_s \times r_s}$, $U_t \in \mathbb{R}^{n_t \times r_t}$, $U_x \in \mathbb{R}^{n_x \times r_x}$ that solve

$$\min_{\mathcal{G}, U_s, U_t, U_x} \|\mathcal{Z} - \mathcal{G} \times_1 U_s \times_2 U_t \times_3 U_x\|,$$

subject to orthonormality of the columns of each factor matrix. Furthermore, note that $\times_k$ denotes the k -mode tensor-times-matrix product. For a tensor $\mathcal{A}$ of size $I_1 \times \cdots \times I_{k-1} \times I_k \times I_{k+1} \times \cdots \times I_N$ and a matrix B of size $J \times I_k$, the k -mode product $\mathcal{A} \times_k B$ contracts the k -th mode of $\mathcal{A}$ against

the columns of B , replacing the k -th dimension I_k with J . The resulting tensor has size $I_1 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_N$, with elementwise definition

$$(\mathcal{A} \times_k B)[i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_N] = \sum_{i_k=1}^{I_k} \mathcal{A}[i_1, i_2, \dots, i_{k-1}, i_k, i_{k+1}, \dots, i_N] \cdot B[j, i_k],$$

where bracket notation indexes tensor elements. Equivalently, this corresponds to multiplying the mode- k unfolding of $\mathcal{A}$ by B and refolding the result. For $N = 2$, it reduces to standard matrix multiplication. For more details and helpful examples, see [Kolda and Bader \[2009\]](#).

The HOOI procedure proceeds as follows:

1. **Initialization:** Estimate each factor matrix U_s, U_t, U_x using Higher-Order Singular Value Decomposition (HOSVD), a generalization of matrix SVD in which the tensor is unfolded along each mode and the leading left singular vectors of each unfolding are used to initialize the corresponding factor matrices.
2. **Iterative updates:** Alternately update each factor matrix while keeping the others fixed. For each mode $n \in \{s, t, x\}$:
 - (a) Compute an intermediate tensor by projecting $\mathcal{Z}$ onto the subspaces spanned by the current estimates of the other factor matrices:

$$\mathcal{Y} = \mathcal{Z} \times_{i \neq n} U_i^\top$$

- (b) Unfold $\mathcal{Y}$ along mode n to form a matrix $\mathcal{Y}_{(n)}$ of size $I_n \times (\prod_{i \neq n} R_i)$.
- (c) Update U_n with the leading r_n left singular vectors of $\mathcal{Y}_{(n)}$.

Repeat this round-robin process until the reconstruction error

$$\|\mathcal{Z} - \mathcal{G} \times_1 U_s \times_2 U_t \times_3 U_x\|_F$$

stabilizes or a maximum number of iterations is reached.

3. **Core tensor:** Once the factor matrices converge, compute

$$\mathcal{G} = \mathcal{Z} \times_1 U_s^\top \times_2 U_t^\top \times_3 U_x^\top.$$

B.2 Components of Tucker Decomposition

The factor matrices capture the dominant patterns along each mode (space, time, design), while the core tensor $\mathcal{G}$ encodes interactions among these patterns. Visualizations of the factor matrices and core tensor are provided in Figures S8 and S9, respectively.

The entries of $\mathcal{G}$ quantify interactions among the columns of the factor matrices (i.e., mode-specific basis vectors). For example, $\mathcal{G}[1, 1, 1] = -29.79$ indicates that the first spatial, temporal, and design bases have a large contribution to the decomposition. In contrast, $\mathcal{G}[1, 2, 1] = -0.59$ represents a weak contribution, indicating that this combination contributes less to the reconstructed tensor.

The spatial and temporal factor matrices are combined via outer product to form a set of $r_s r_t = 4$ separable matrices. Then, as outlined in Section 3.2.1, the core tensor is used to create a linear

combination of these outer product matrices, yielding bases $\mathcal{B}_i \in \mathbb{R}^{n_s \times n_t}$, $i = 1, \dots, r_x$.¹ See Figure S10 for visualization of bases $\mathcal{B}_1$ and $\mathcal{B}_2$ corresponding to the example data.

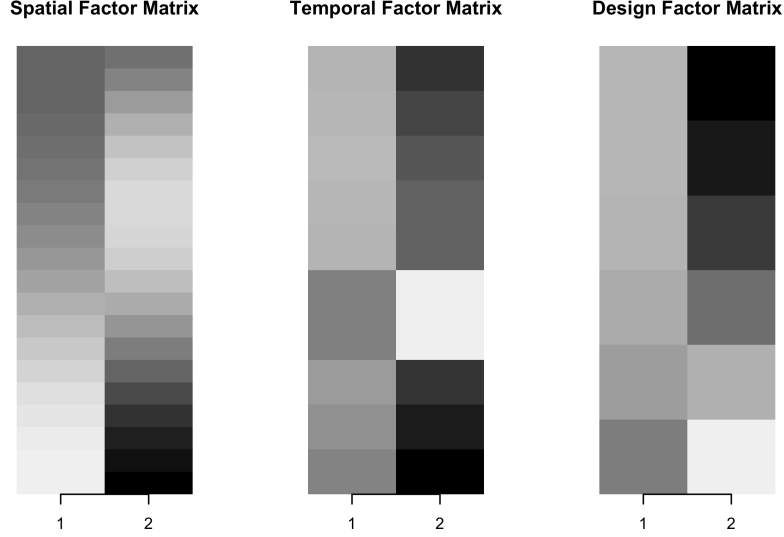


Figure S8: Factor matrices U_s , U_t , and U_x capturing dominant patterns along each mode.

See Figure S11 for visualization of the four-dimensional Tucker decomposition.

To illustrate how the spatial bases are utilized at different temporal points, we compute the relative importance of each spatial basis $U_s[:, j_s]$ directly from the factor matrices and core tensor. Specifically, for spatial basis j_s , we define its importance at time t as

$$\text{importance}(j_s, t) = \sum_{j_t=1}^{r_t} U_t[t, j_t]^2 \sum_{i=1}^{r_x} \mathcal{G}[j_s, j_t, i]^2,$$

which quantifies how strongly this spatial basis interacts with all temporal bases and design components at time t . Each squared core entry $\mathcal{G}[j_s, j_t, i]^2$ is the time-independent energy of the rank-one component pairing spatial basis j_s with temporal basis j_t and design basis i ; under orthonormal factor matrices these squared entries partition the total variance retained by the decomposition [Sidiropoulos et al., 2017]. The weight $U_t[t, j_t]^2$ measures how active temporal basis j_t is at time t : since the columns of U_t are orthonormal, $\sum_{j_t'} U_t[t, j_t']^2 = 1$, so $U_t[t, j_t]^2$ is the fraction of that temporal basis's activity occurring at time t . The product therefore allocates each component's energy to time t in proportion to its temporal basis's activity there, and summing over j_t and i gives the total energy attributed to spatial basis j_s at time t . Because the core couples each spatial basis to particular temporal bases, and different temporal bases are active at different times, this contribution shifts across the temporal domain, revealing when each spatial pattern is most prominent. Summing over all time points returns the total energy of spatial basis j_s , so the measure distributes that total across time. Normalizing across spatial bases yields relative importance (RI),

$$\text{RI}(j_s, t) = \frac{\text{importance}(j_s, t)}{\sum_{j_s'=1}^{r_s} \text{importance}(j_s', t)}.$$

¹In this example, the spatiotemporal bases $\mathcal{B}$ are matrices, since they are formed as outer products of spatial and temporal basis vectors. In our manuscript, the bases $\mathcal{B}$ are three-dimensional tensors because the temporal dimension is split into month and year.

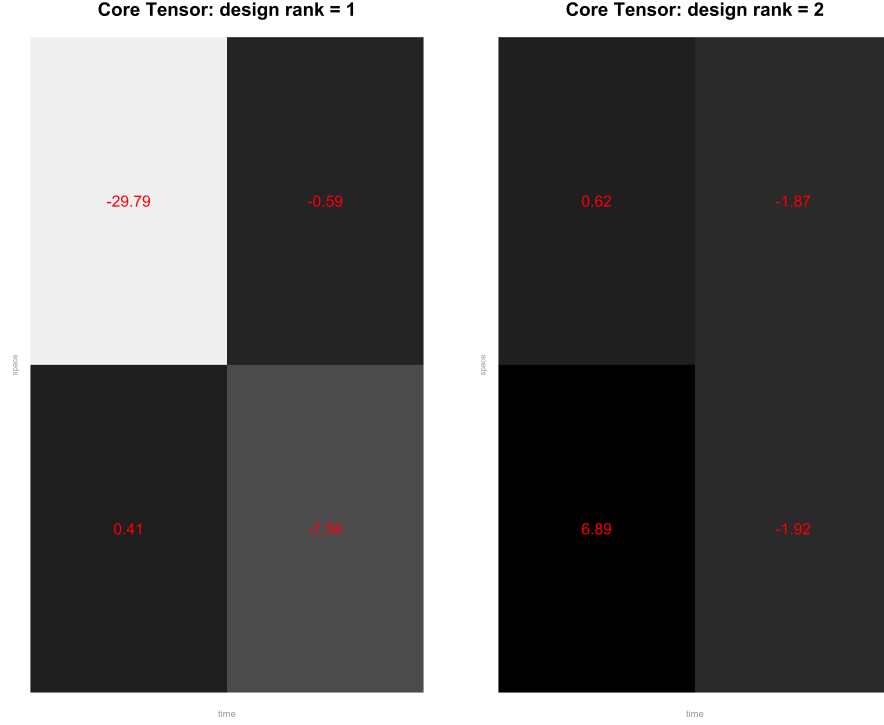


Figure S9: Core tensor $\mathcal{G} \in \mathbb{R}^{2 \times 2 \times 2}$, shown as slices along the last dimension, with values in red.

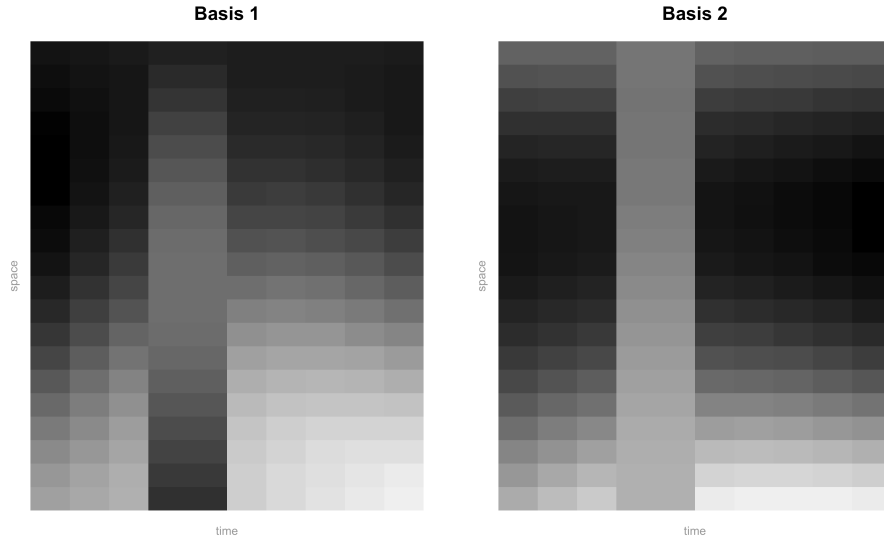


Figure S10: Spatiotemporal basis matrices $\mathcal{B}_1$ and $\mathcal{B}_2$.

Unlike the aggregate proportion of variance reported in Section 5, which is summed over all time points, this measure is resolved over time, revealing how the contribution of each spatial basis shifts across the temporal domain.

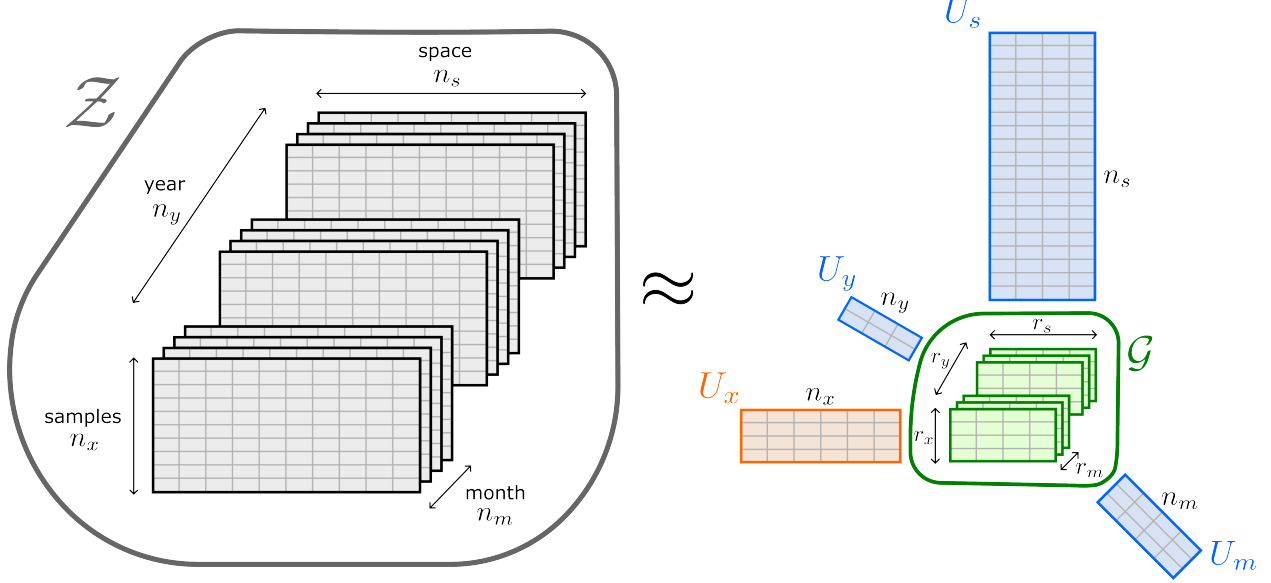


Figure S11: Illustration of the 4D Tucker decomposition of the simulator output tensor $\mathcal{Z} \in \mathbb{R}^{n_s \times n_m \times n_y \times n_x}$, where the four modes correspond to space (n_s), month (n_m), year (n_y), and design samples (n_x). The tensor is approximated by a core tensor $\mathcal{G} \in \mathbb{R}^{r_s \times r_m \times r_y \times r_x}$ (green) and four factor matrices: the spatial factor matrix $U_s \in \mathbb{R}^{n_s \times r_s}$, the monthly factor matrix $U_m \in \mathbb{R}^{n_m \times r_m}$, and the yearly factor matrix $U_y \in \mathbb{R}^{n_y \times r_y}$ (all in blue), and the design factor matrix $U_x \in \mathbb{R}^{n_x \times r_x}$ (orange). Each factor matrix extracts the dominant patterns along its respective mode, while the core tensor $\mathcal{G}$ weights the interactions among these mode-specific bases. The ranks (r_s, r_m, r_y, r_x), shown explicitly on the core tensor, control the degree of compression along each mode, with smaller ranks yielding a more parsimonious representation.

For example, at time points $t = 1, 5, 10$, the normalized importances are

$$\begin{aligned} t = 1 : & \quad (0.74, 0.26), \\ t = 5 : & \quad (0.58, 0.42), \\ t = 10 : & \quad (0.94, 0.06), \end{aligned}$$

where each pair corresponds to spatial bases 1 and 2, respectively. These results indicate that different spatial patterns dominate at different times. For instance, the first spatial basis dominates early and late time points ($t = 1, 10$), while a mixture of the two bases is more prominent in the middle of the temporal domain ($t = 5$).

C Mathematical Derivations

C.1 Single-Fidelity Posterior

In Section 3.2.2, the full model in (7) is re-expressed as (8). In this section, we obtain that result. First, we restate the hierarchical model expressed in (7):

$$\begin{aligned} \mathbf{z} \mid \mathbf{w}, \lambda_\eta &\sim \mathcal{N}(B\mathbf{w}, \lambda_\eta^{-1}I) \\ \mathbf{w} &\sim \mathcal{N}(\mathbf{0}, \Sigma_w(\mathcal{X}, \mathcal{X})) \\ \lambda_\eta &\sim \Gamma(a_\eta, b_\eta). \end{aligned}$$

Observe that

$$\begin{aligned} p(\mathbf{z} \mid \mathbf{w}, \lambda_\eta) \times p(\lambda_\eta \mid a, b) &\propto |\lambda_\eta^{-1}I|^{-1/2} \exp \left\{ -\frac{1}{2} (\mathbf{z} - B\mathbf{w})^\top (\lambda_\eta^{-1}I)^{-1} (\mathbf{z} - B\mathbf{w}) \right\} \times \lambda_\eta^{a-1} \exp \{-b\lambda_\eta\} \\ &:= c \times \exp \left\{ -\frac{1}{2} (\hat{\mathbf{w}} - \mathbf{w})^\top \Sigma_*^{-1} (\hat{\mathbf{w}} - \mathbf{w}) \right\}, \end{aligned}$$

where c is a value that does not involve $\mathbf{w}$. By completing the square, we obtain

$$\begin{aligned} \hat{\mathbf{w}} &= \left(B^\top B \right)^{-1} B^\top \mathbf{z} = \text{vec}(U_x), \\ \Sigma_* &= \lambda_\eta^{-1} \left(B^\top B \right)^{-1}, \\ c &= \lambda_\eta^{\frac{n_x n}{2} + a - 1} \exp \left\{ -\lambda_\eta \left[\frac{1}{2} \mathbf{z}^\top \left(I - B(B^\top B)^{-1} B^\top \right) \mathbf{z} + b \right] \right\} \\ &:= \lambda_\eta^{\frac{n_x r_x}{2}} \times \lambda_\eta^{a' - 1} \exp \{-\lambda_\eta b'\}, \end{aligned}$$

where $\mathbf{z} \in \mathbb{R}^{n_x n}$ and the factor $\lambda_\eta^{n_x r_x / 2}$ is the normalizing constant $|\Sigma_*|^{-1/2}$ of the $\hat{\mathbf{w}}$ density, so that

$$a' = a + \frac{n_x(n - r_x)}{2}, \quad b' = b + \frac{1}{2} \mathbf{z}^\top \left(I - B(B^\top B)^{-1} B^\top \right) \mathbf{z},$$

matching a'_η and b'_η in Section 3.2.2 of the main text. Therefore,

$$p(\mathbf{z} \mid \mathbf{w}, \lambda_\eta) \times p(\lambda_\eta \mid a, b) \propto p(\hat{\mathbf{w}} \mid \mathbf{w}, \lambda_\eta) \times p(\lambda_\eta \mid a', b').$$

This yields the hierarchical model

$$\begin{aligned} \hat{\mathbf{w}} \mid \mathbf{w}, \lambda_\eta &\sim \mathcal{N}(\mathbf{w}, \lambda_\eta^{-1}(B^\top B)^{-1}) \\ \mathbf{w} &\sim \mathcal{N}(\mathbf{0}, \Sigma_w(\mathcal{X}, \mathcal{X})) \\ \lambda_\eta &\sim \Gamma(a', b'). \end{aligned}$$

Next, we integrate out $\mathbf{w}$, starting with

$$\begin{aligned} p(\hat{\mathbf{w}} \mid \lambda_\eta) &= \int p(\hat{\mathbf{w}}, \mathbf{w} \mid \lambda_\eta) d\mathbf{w} \\ &= \int p(\hat{\mathbf{w}} \mid \mathbf{w}, \lambda_\eta) \times p(\mathbf{w}) d\mathbf{w} \\ &\propto \int \exp \left\{ -\frac{1}{2} (\hat{\mathbf{w}} - \mathbf{w})^\top (\lambda_\eta B^\top B) (\hat{\mathbf{w}} - \mathbf{w}) \right\} \times \exp \left\{ -\frac{1}{2} \mathbf{w}^\top \Sigma_w^{-1} \mathbf{w} \right\} d\mathbf{w} \\ &= \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top (\lambda_\eta B^\top B) \hat{\mathbf{w}} \right\} \times \int \exp \left\{ -\frac{1}{2} \left[\mathbf{w}^\top (\lambda_\eta B^\top B + \Sigma_w^{-1}) \mathbf{w} - 2\mathbf{w}^\top (\lambda_\eta B^\top B) \hat{\mathbf{w}} \right] \right\} d\mathbf{w}. \end{aligned}$$

Let

$$Q = \lambda_\eta^{-1}(B^\top B)^{-1}, \quad R = \Sigma_w(\mathcal{X}, \mathcal{X}), \quad S = Q^{-1} + R^{-1}.$$

By completing the square, we obtain

$$\begin{aligned} & \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top Q^{-1} \hat{\mathbf{w}} \right\} \times \exp \left\{ \frac{1}{2} \hat{\mathbf{w}}^\top (Q^{-1} S^{-1} Q^{-1}) \hat{\mathbf{w}} \right\} \times \int \exp \left\{ -\frac{1}{2} (\mathbf{w} - S^{-1} Q^{-1} \hat{\mathbf{w}})^\top S (\mathbf{w} - S^{-1} Q^{-1} \hat{\mathbf{w}}) \right\} d\mathbf{w} \\ & \propto \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top Q^{-1} \hat{\mathbf{w}} \right\} \times \exp \left\{ \frac{1}{2} \hat{\mathbf{w}}^\top (Q^{-1} S^{-1} Q^{-1}) \hat{\mathbf{w}} \right\} \\ & = \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top [Q^{-1} - Q^{-1}(Q^{-1} + R^{-1})^{-1} Q^{-1}] \hat{\mathbf{w}} \right\} \\ & = \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top (Q + R)^{-1} \hat{\mathbf{w}} \right\} \tag{*} \\ & = \exp \left\{ -\frac{1}{2} \hat{\mathbf{w}}^\top [\lambda_\eta^{-1}(B^\top B)^{-1} + \Sigma_w(\mathcal{X}, \mathcal{X})]^{-1} \hat{\mathbf{w}} \right\} \end{aligned}$$

where $(*)$ follows from the Sherman–Morrison–Woodbury matrix identity. Therefore, we have successfully integrated out $\mathbf{w}$, yielding a model that matches (8):

$$\begin{aligned} \hat{\mathbf{w}} \mid \lambda_\eta & \sim \mathcal{N} \left(\mathbf{0}, \lambda_\eta^{-1} (B^\top B)^{-1} + \Sigma_w(\mathcal{X}, \mathcal{X}) \right) \\ \lambda_\eta & \sim \Gamma(a', b'). \end{aligned}$$

C.2 Multi-Fidelity Posterior

In Section 3.3.4, we reduce the hierarchical model in (10) to that of (11). Many of the steps are analogous to Section C.1. Begin with the full model,

$$\begin{aligned} \begin{bmatrix} \mathbf{z}^{\text{LF}} \\ \mathbf{z}^{\text{HF}} \end{bmatrix} \mid \boldsymbol{\theta}, \lambda_\eta, \lambda_\delta & \sim \mathcal{N} \left(\begin{bmatrix} B\mathbf{w}^{\text{LF}} \\ \tilde{B}\mathbf{w}^{\text{HF}} + D\mathbf{v} \end{bmatrix}, \begin{bmatrix} \lambda_\eta^{-1} I & \mathbf{0} \\ \mathbf{0} & \lambda_\delta^{-1} I \end{bmatrix} \right) \\ \boldsymbol{\theta} & \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta) \\ \lambda_\eta & \sim \Gamma(a_\eta, b_\eta), \quad \lambda_\delta \sim \Gamma(a_\delta, b_\delta), \end{aligned}$$

where $\boldsymbol{\theta} = [\mathbf{w}^{\text{LF}\top}, \mathbf{w}^{\text{HF}\top}, \mathbf{v}^\top]^\top$ and

$$\Sigma_\theta = \begin{bmatrix} \Sigma_w(\mathcal{X}^{\text{LF}}, \mathcal{X}^{\text{LF}}) & \Sigma_w(\mathcal{X}^{\text{LF}}, \mathcal{X}^{\text{HF}}) & \mathbf{0} \\ \Sigma_w(\mathcal{X}^{\text{HF}}, \mathcal{X}^{\text{LF}}) & \Sigma_w(\mathcal{X}^{\text{HF}}, \mathcal{X}^{\text{HF}}) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \Sigma_v(\mathcal{X}^{\text{HF}}, \mathcal{X}^{\text{HF}}) \end{bmatrix}.$$

Next, reduce $\mathbf{z}^{\text{LF}} \rightarrow \hat{\mathbf{w}}^{\text{LF}}$ and $\mathbf{z}^{\text{HF}} \rightarrow \hat{\mathbf{w}}^{\text{HF}}, \hat{\mathbf{v}}$, which can be done separately given their independence conditional on $\boldsymbol{\theta}$. This process is the same as the first step in Section C.1, resulting in

$$\begin{aligned} \hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta}, \lambda_\eta, \lambda_\delta & \sim \mathcal{N}(\boldsymbol{\theta}, \Sigma_{\hat{\boldsymbol{\theta}}|\boldsymbol{\theta}}) \\ \boldsymbol{\theta} & \sim \mathcal{N}(\mathbf{0}, \Sigma_\theta) \\ \lambda_\eta & \sim \Gamma(a'_\eta, b'_\eta), \quad \lambda_\delta \sim \Gamma(a'_\delta, b'_\delta) \end{aligned}$$

where $\hat{\boldsymbol{\theta}} = [\hat{\mathbf{w}}^{\text{LF}\top}, \hat{\mathbf{w}}^{\text{HF}\top}, \hat{\mathbf{v}}^\top]^\top$, $K = [\tilde{B} \ D]$, and

$$\Sigma_{\hat{\boldsymbol{\theta}}|\boldsymbol{\theta}} = \begin{bmatrix} \lambda_\eta^{-1} (B^\top B)^{-1} & \mathbf{0} \\ \mathbf{0} & \lambda_\delta^{-1} (K^\top K)^{-1} \end{bmatrix}.$$

Furthermore, the priors for λ_η and λ_δ get updated with

$$\begin{aligned} a'_\eta &= a_\eta + \frac{n_x^{\text{LF}}(n^{\text{LF}} - r_x)}{2}, & b'_\eta &= b_\eta + \frac{1}{2}(\mathbf{z}^{\text{LF}})^\top \left(I - B(B^\top B)^{-1} B^\top \right) \mathbf{z}^{\text{LF}}, \\ a'_\delta &= a_\delta + \frac{n_x^{\text{HF}}[n^{\text{HF}} - (r_x + r'_x)]}{2}, & b'_\delta &= b_\delta + \frac{1}{2}(\mathbf{z}^{\text{HF}})^\top \left(I - K(K^\top K)^{-1} K^\top \right) \mathbf{z}^{\text{HF}}. \end{aligned}$$

We next integrate out $\boldsymbol{\theta}$, which is analogous to the second step in Section C.1:

$$\begin{aligned} p(\hat{\boldsymbol{\theta}} \mid \lambda_\eta, \lambda_\delta) &= \int p(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta}, \lambda_\eta, \lambda_\delta) \times p(\boldsymbol{\theta}) d\boldsymbol{\theta} \\ &= \int \mathcal{N}(\boldsymbol{\theta}, \Sigma_{\hat{\boldsymbol{\theta}}|\boldsymbol{\theta}}) \times \mathcal{N}(\mathbf{0}, \Sigma_\theta) d\boldsymbol{\theta} \\ &\propto \mathcal{N}(\mathbf{0}, \Sigma_{\hat{\boldsymbol{\theta}}|\boldsymbol{\theta}} + \Sigma_\theta). \end{aligned}$$

By performing this integration, we obtain the model that is expressed in (11):

$$\begin{aligned} \hat{\boldsymbol{\theta}} \mid \lambda_\eta, \lambda_\delta &\sim \mathcal{N}(\mathbf{0}, \Sigma_{\hat{\boldsymbol{\theta}}}) \\ \lambda_\eta &\sim \Gamma(a'_\eta, b'_\eta), \quad \lambda_\delta \sim \Gamma(a'_\delta, b'_\delta), \end{aligned}$$

where $\Sigma_{\hat{\boldsymbol{\theta}}} := \Sigma_{\hat{\boldsymbol{\theta}}|\boldsymbol{\theta}} + \Sigma_\theta$. Though not shown, there are hyperparameters corresponding to the precision and length scales of the GP priors for the weights. See Section E.4 for details on the hyperpriors.

Using this hierarchical model, we obtain the posterior

$$p(\lambda_\eta, \lambda_\delta \mid \hat{\boldsymbol{\theta}}) \propto p(\hat{\boldsymbol{\theta}} \mid \lambda_\eta, \lambda_\delta) \times p(\lambda_\eta) \times p(\lambda_\delta),$$

which is sampled via Markov chain Monte Carlo.

C.3 Matrix Calculation

In Section 3.2.1, the matrix $B \in \mathbb{R}^{(n_x n) \times (n_x r_x)}$ becomes large when there are many spatiotemporal locations (i.e., large n). However, we do not need to directly construct B to compute the quantities $(B^\top B)^{-1}$ or $B^\top \mathbf{z}$ that are required for (8); we exploit the sparse definition of B to compute these terms more efficiently. By the mixed product property of Kronecker products, B has the form

$$B = \begin{bmatrix} \mathbf{c}_1 & & & \mathbf{c}_{r_x} & & \\ & \ddots & & & \ddots & \\ & & \mathbf{c}_1 & & & \\ & & & \dots & & \\ & & & & \ddots & \\ & & & & & \mathbf{c}_{r_x} \end{bmatrix},$$

where $\mathbf{c}_k \in \mathbb{R}^n$ is

$$\mathbf{c}_k = \sum_{j_s=1}^{r_s} \sum_{j_m=1}^{r_m} \sum_{j_y=1}^{r_y} \mathbf{b}_{j_s j_m j_y} G_{(4)}^\top [(j_s - 1)r_m r_y + (j_m - 1)r_y + j_y, k], \quad k = 1, 2, \dots, r_x.$$

Therefore, the information in B can be distilled into the vectors $\mathbf{c}_k$. Then, we compute the block matrix

$$B^\top B = \begin{bmatrix} \begin{matrix} c_{11} & & \\ & \ddots & \\ & & c_{11} \end{matrix} & \cdots & \begin{matrix} c_{1r_x} & & \\ & \ddots & \\ & & c_{1r_x} \end{matrix} \\ \vdots & \ddots & \vdots \\ \begin{matrix} c_{r_x 1} & & \\ & \ddots & \\ & & c_{r_x 1} \end{matrix} & \cdots & \begin{matrix} c_{r_x r_x} & & \\ & \ddots & \\ & & c_{r_x r_x} \end{matrix} \end{bmatrix}$$

where each $n_x \times n_x$ gray block is diagonal with elements $c_{mn} = \mathbf{c}_m^\top \mathbf{c}_n$ for $1 \leq m, n \leq r_x$. Let $C_{\text{coef}} \in \mathbb{R}^{r_x \times r_x}$ be the matrix whose entries are c_{mn} . Then

$$B^\top B = C_{\text{coef}} \otimes I_{n_x}.$$

Using the Kronecker inverse identity,

$$(B^\top B)^{-1} = C_{\text{coef}}^{-1} \otimes I_{n_x},$$

which is computationally more efficient to compute, only requiring inversion of an $r_x \times r_x$ matrix.

To compute $B^\top \mathbf{z}$, we have

$$B^\top \mathbf{z} = [\mathbf{c}_1^\top \mathbf{z}_1 \quad \cdots \quad \mathbf{c}_1^\top \mathbf{z}_{n_x} \quad \cdots \quad \mathbf{c}_{r_x}^\top \mathbf{z}_1 \quad \cdots \quad \mathbf{c}_{r_x}^\top \mathbf{z}_{n_x}]^\top,$$

where $\mathbf{z}_i$ collects the vectorized spatiotemporal outputs for the i -th design point, for $i = 1, \dots, n_x$. Therefore, we have found $(B^\top B)^{-1}$ and $B^\top \mathbf{z}$ without computing B directly.

In Section 3.3.4, the matrix $K = [\tilde{B} \quad D]$ has a similar structure to B :

$$K = \begin{bmatrix} \tilde{\mathbf{c}}_1 & & \tilde{\mathbf{c}}_{r_x} & & \mathbf{c}'_1 & & \mathbf{c}'_{r'_x} \\ & \ddots & & \ddots & & \ddots & \\ & & \tilde{\mathbf{c}}_1 & & \mathbf{c}'_1 & & \\ & & & \tilde{\mathbf{c}}_{r_x} & & & \\ & & & & \mathbf{c}'_1 & & \\ & & & & & \ddots & \\ & & & & & & \mathbf{c}'_{r'_x} \end{bmatrix},$$

where

$$\tilde{\mathbf{c}}_k = \sum_{j_s=1}^{r_s} \sum_{j_m=1}^{r_m} \sum_{j_y=1}^{r_y} \tilde{\mathbf{b}}_{j_s j_m j_y} G_{(4)}^\top [(j_s - 1)r_m r_y + (j_m - 1)r_y + j_y, k], \quad k = 1, 2, \dots, r_x,$$

and

$$\mathbf{c}'_k = \sum_{j'_s=1}^{r'_s} \sum_{j'_m=1}^{r'_m} \sum_{j'_y=1}^{r'_y} \mathbf{d}_{j'_s j'_m j'_y} G_{(4)}'^\top [(j'_s - 1)r'_m r'_y + (j'_m - 1)r'_y + j'_y, k], \quad k = 1, 2, \dots, r'_x.$$

Therefore, $(K^\top K)^{-1}$ and $K^\top \mathbf{z}^{\text{HF}}$ can be calculated analogously to the above derivations.

D Simulation Study Details

This section gives the design rationale for the simulation study of Section 4, the analytical forms of the low-fidelity simulator and of the discrepancy that produces the high-fidelity output, and the ranks used by the tensor-based methods.

Simulations were run on 25×25 (LF) and 50×50 (HF) spatial grids with monthly outputs over a five-year period, using $n_x^{\text{LF}} = 100$ low-fidelity and $n_x^{\text{HF}} = 10$ high-fidelity design points, and were evaluated at 100 held-out test inputs. The study was designed around two principles. First, we induced a significant LF–HF discrepancy so that the LF simulator, even after interpolation to the fine grid, could not reliably predict unobserved HF outputs; this ensured the discrepancy term was necessary and that the multi-fidelity structure was being tested rather than a simple rescaling of the LF emulator. Second, we ensured adequate input-space variability in the HF response so that the sparsely sampled HF data alone ($n_x^{\text{HF}} = 10$) were insufficient to recover the full model behavior, requiring genuine information transfer from the LF ensemble. In addition, the third input x_3 was constructed as a dummy variable that did not affect the LF or HF outputs, allowing us to verify that the emulator correctly identifies irrelevant inputs through its estimated length scales.

The low-fidelity simulator is based on the Shubert function [Surjanovic and Bingham, 2013] and does not impose a separable structure. It is defined as follows, where **d1** and **d2** are the active design inputs x_1 and x_2 ; the dummy input x_3 does not appear, since by construction it has no effect on the output:

```

1  shubert <- Vectorize(function(
2      s1, s2,    # spatial coordinates
3      m,        # month
4      y,        # year
5      d1, d2,    # design parameters
6      start, end # Shubert start/end indices
7  ) {
8      ii <- start:end
9
10     alpha <- 1 + sin(y) / 2 +
11     (1/10) * (sin(s1 * (d1/2 + 0.5)) + cos(s2 * (d2/2 + 0.5))) * sin(m)
12     beta  <- (1/10) * (s1 + s2) * cos(m) + (1/20) * sin(m) + (1/5) * y
13
14     sum1 <- sum(ii * cos((ii + 1) * d1 * alpha + ii + beta))
15     sum2 <- sum(ii * cos((ii + 1) * d2 * alpha + ii + beta))
16
17     sum1 + sum2 + (1/5) * y
18 })

```

While **alpha** and **beta** (lines 10–12) are composed of separable spatial and temporal components, **sum1** and **sum2** introduce nonseparability by applying a nonlinear cosine transformation to expressions that combine space and time. Specifically, the terms

$$\sum_i i \cos[(i+1)d_k \alpha(s_1, s_2, m, y, d_1, d_2) + i + \beta(s_1, s_2, m, y)], \quad k \in \{1, 2\},$$

mix space, time, and the design inputs inside a nonlinear function, preventing factorization into purely spatial and temporal components. The start and end terms, of which we utilized **start** = 1 and **end** = 5, define the number of terms summed over, and therefore control the complexity and oscillatory behavior of the resulting function.

To calculate the HF data, we create a discrepancy that gets added to the (interpolated) LF data.

```

1   sigma_val <- abs(sin(mo_cyclic) + cos(mo_cyclic) * yr)
2   Sigma <- sigma_val * diag(2)
3   M_disc[, mo, yr] <- dmvnorm(coord_hf, mean = c(0, 0), sigma = Sigma)

```

Here `mo_cyclic` and `yr` are the cyclic month and year encodings, `coord_hf` holds the coordinates of the high-fidelity grid, `M_disc` is the resulting discrepancy field, and `dmvnorm` is the multivariate normal density from the `mvtnorm` package. The discrepancy adds a positive “bump” to the interpolated low-fidelity field, given by the density of a bivariate normal distribution with fixed center and isotropic covariance $\sigma^2(m, y)I_2$, so that only its width varies with month and year. Therefore, while the low-fidelity data depend on `d1` and `d2` in a complicated way, the discrepancy depends only on the month and year. This was a purposeful choice for the simulation study, so few high-fidelity runs are needed to understand the discrepancy.

When implementing tensor-based methods, we require ranks (4, 4, 3, 11) for the low-fidelity data and ranks (18, 8, 4, 8) for the discrepancy to capture 95% explained variance.

E Emulator Implementation Details

E.1 Compared Emulators

We compared seven emulators, using the names introduced in Section 4 of the main text.

1. **LF-TEN**: a single-fidelity tensor emulator trained on LF data and then interpolated to the fine grid, following Section 3.2.
2. **HF-TEN**: a single-fidelity tensor emulator trained on HF data, following Section 3.2.
3. **MF-TEN**: the multi-fidelity tensor emulator combining LF and HF data, following Section 3.3.
4. **MF-PCA**: a multi-fidelity emulator using principal component-based dimension reduction applied to vectorized spatiotemporal outputs (stacking space and time into a single vector). This yields a low-rank nonseparable basis representation but lacks the flexible cross-mode interactions captured by Tucker decomposition. The number of ranks was selected to explain the same percentage of variance as the Tucker decomposition in MF-TEN, extending the approach of Higdon et al. [2008] to high-dimensional spatiotemporal and multi-fidelity emulation.
5. **MF-SEP**: a multi-fidelity emulator that models the full output directly, without a low-rank basis. Following the additive structure, the HF output is modeled as $\eta^{\text{HF}} = \eta^{\text{LF}} + \delta$, where η^{LF} and δ are independent zero-mean Gaussian processes, each with its own covariance separable across space, month, year, and design,

$$\Sigma = \sigma^2 (\Sigma_s \otimes \Sigma_m \otimes \Sigma_y \otimes \Sigma_x).$$

Squared-exponential correlation kernels with automatic relevance determination (ARD) are used for space, year, and design, and a periodic kernel $\exp\{-2\sin^2(\pi d/12)/\ell_m\}$ of period 12 for month, where d is the difference in month index; each factor has unit diagonal, so that σ^2 carries the overall scale. Because the HF design points are a subset of the LF design points, the joint model decomposes exactly into independent Gaussian processes for the LF output and for the HF–LF residual at the HF design points, so the two components are fit separately and their predictions combined additively. This split into independent LF and residual components holds when both fidelities are observed on a common spatial grid; for MPAS-Seaice, where they are not, the LF runs are first interpolated onto the HF mesh, as described at the end of this section.

Prediction is carried out on the same space–month–year grid used for training, so the design is the only mode in which prediction at new values occurs. The spatiotemporal factors then cancel from the kriging predictor, and the predictive mean of each component is

$$\left\{ I \otimes \left[\Sigma_x(\mathcal{X}, \mathbf{x}^*)^\top \Sigma_x(\mathcal{X}, \mathcal{X})^{-1} \right] \right\} \mathbf{z},$$

where $\mathbf{z}$ denotes that component’s training outputs (the interpolated LF runs at the LF design points, or the HF–LF residuals at the HF design points) and $\Sigma_x(\mathcal{X}, \mathbf{x}^*)$, $\Sigma_x(\mathcal{X}, \mathcal{X})$ are formed from the corresponding design, resulting in a weighted combination of the training runs, with weights that depend only on the design input. MF-SEP is therefore linear in the output, like MF-TEN, but imposes no rank truncation. The predictive covariance of each

component factorizes as

$$\sigma^2 \left\{ 1 - \Sigma_x(\mathcal{X}, \mathbf{x}^*)^\top \Sigma_x(\mathcal{X}, \mathcal{X})^{-1} \Sigma_x(\mathcal{X}, \mathbf{x}^*) \right\} (\Sigma_s \otimes \Sigma_m \otimes \Sigma_y),$$

in which only the scalar prefactor depends on the test input. The marginal predictive variance for $\mathbf{x}^*$ is therefore that same scalar at every spatiotemporal location, but the off-diagonal entries are nonzero, so MF-SEP retains a full spatiotemporal correlation implied by its fitted kernels.

We estimate the length scales by coordinate-wise grid search on the negative log-likelihood and plug the estimates in for prediction. In the simulation study, the spatial grid is small enough for the likelihood to be evaluated exactly. In MPAS-Seaice, however, Σ_s is $33,530 \times 33,530$, too large to store or invert, so we replace it with a Vecchia approximation [Vecchia, 1988] in which each cell centroid is conditioned on its 30 nearest predecessors under a maximin ordering.

When the fidelities are not observed on a common spatial grid, as in MPAS-Seaice, the LF process is not observed at the HF locations, and the exact model would kriging it onto the HF grid within the joint likelihood. The HF-LF residual would then combine the discrepancy with the LF kriging error, its covariance would become a sum of two Kronecker products rather than a single one, and the spatial covariances would no longer cancel from the predictive mean. Evaluating this likelihood exactly requires dense operations on the HF spatial covariance, which is infeasible at the resolution of the MPAS-Seaice mesh. A Vecchia approximation could in principle make it tractable, but the Vecchia factors of the individual spatial covariances cannot simply be combined, because the inverse of a sum of Kronecker products does not inherit their sparsity. The approximation would instead have to be applied to the residual process directly, whose covariance entries depend on the LF kriging error and must be recomputed at every likelihood evaluation during hyperparameter estimation, which requires a custom implementation beyond the scope of this paper. We therefore first interpolate each LF run onto the HF mesh by inverse-distance weighting over the $k = 4$ nearest LF cells, a computationally inexpensive substitute for kriging, and fit MF-SEP to the interpolated LF outputs. This reduces the problem to a common grid, on which the exact decomposition applies. The cost is that the interpolation error is not modeled separately; it is absorbed into the discrepancy process and modeled with the discrepancy’s covariance, and the uncertainty of the interpolation itself is not propagated, so MF-SEP’s predictive uncertainty is conditional on the interpolated LF fields.

6. **MF-IDP**: a baseline that applies the Kennedy and O’Hagan [2000] multi-fidelity construction independently at each spatiotemporal location, and therefore carries no correlation structure across space or time. Because a separate model is fit at each location, each location requires both an LF and an HF value. Since MF-IDP has no spatial model with which to kriging the LF process onto the HF mesh, the LF ensemble for MPAS-Seaice is interpolated beforehand, using the same inverse-distance weighting over the $k = 4$ nearest LF cells as for MF-SEP. As with MF-SEP, the interpolation error is not propagated, so MF-IDP’s predictive uncertainty is conditional on the interpolated LF fields.
7. **MF-FFN**: a feed-forward neural network emulator that pools the LF and HF observations into a single training set, distinguished by a binary fidelity indicator. Each row is described by the spatial coordinates, month, year, the fidelity indicator, and the design inputs. Month enters through a cyclical sin/cos encoding of period 12, and each spatial coordinate is ex-

panded into a deterministic Fourier feature bank $\{\sin(2\pi ks/L), \cos(2\pi ks/L)\}_{k=1}^{10}$ over the coordinate span L . Such sinusoidal encodings let a network represent high-frequency spatial structure that it otherwise learns only slowly [Tancik et al., 2020], and have been used to encode geographic position [Rußwurm et al., 2024]; here the harmonics are fixed rather than randomly drawn.

The network has two fully connected hidden layers of 128 and 64 ReLU units and two output heads returning a predictive mean and log-variance, with the mean passed through a scaled sigmoid so that predictions respect the $[0, 1]$ bound on sea-ice concentration. It is trained by minimizing the continuous ranked probability score, a proper scoring rule that, unlike the Gaussian negative log-likelihood, does not reward inflating the predicted variance to absorb a large residual. The score is evaluated in closed form by treating the output at each location as Gaussian with the predicted mean and variance [Gneiting and Raftery, 2007]. Predictive uncertainty is then calibrated by normalized split conformal prediction [Lei et al., 2018] on held-out HF runs, which rescales the network’s predictive standard deviation by a factor chosen so that the interval of ± 1.96 rescaled standard deviations around the predictive mean attains the nominal 95% coverage on those runs.

Split conformal requires a calibration set disjoint from the data used to fit the network, so within each cross-validation fold the available high-fidelity design points are partitioned into a fitting set and a calibration set, and neither contains the held-out run. No information from the test design point enters the fit or the calibration. The calibration set is necessarily small, so the attainable coverage levels are coarse, and MF-FFN’s intervals should be read as approximately rather than exactly calibrated.

We include MF-FFN to contrast the linear basis representation underlying MF-TEN with a nonlinear alternative, rather than as an exhaustive comparison against neural network emulators. Multi-fidelity neural networks for spatiotemporal output remain an active area of research, and a fuller assessment of that class of methods is beyond the scope of this work.

Two aspects of this implementation work in MF-FFN’s favor relative to the other emulators, and we note them so that the comparison is read correctly. First, CRPS is both the training objective for MF-FFN and one of the four evaluation metrics reported in Sections 4 and 5. None of the remaining emulators is fit to a criterion on which it is subsequently scored, since the tensor- and PCA-based methods target the posterior and MF-SEP and MF-IDP are likelihood-based. Second, the conformal step calibrates MF-FFN’s predictive intervals to the nominal 95% level by construction, so its coverage is close to nominal largely by design rather than as a consequence of the fitted model. Therefore, on CRPS and coverage, the comparison is tilted in MF-FFN’s favor by construction rather than by what the fitted model achieves.

The three tensor-based methods (LF-TEN, HF-TEN, MF-TEN) use Tucker decomposition to construct low-dimensional bases and MF-PCA uses an SVD of the vectorized data matrix; the remaining three methods (MF-SEP, MF-IDP, MF-FFN) do not involve such basis decomposition.

For both MF-SEP and MF-IDP, the interpolated LF field is treated as if it had been observed exactly. The error incurred by the interpolation is not propagated into the predictive distribution, so the uncertainty reported for these two methods is conditional on the interpolant and, to that extent, understates the total uncertainty.

E.2 Evaluation Metrics

Predictions at each held-out test input are compared to the true high-fidelity output using three marginal metrics, computed independently at each spatiotemporal location and then averaged, and one joint metric computed over the full spatiotemporal field.

Let z denote the true output at a given location, and $\hat{\mu}$ and $\hat{\sigma}$ the predictive mean and standard deviation there. The absolute error (AE) is $|z - \hat{\mu}|$. Coverage is the proportion of locations at which z falls inside the nominal 95% predictive interval $\hat{\mu} \pm 1.96\hat{\sigma}$. The continuous ranked probability score (CRPS) [Gneiting and Raftery, 2007] is computed from the predictive sample itself rather than from a closed-form Gaussian expression, so that it remains valid for methods whose predictive distribution is not normal. Writing $x_1, \dots, x_M$ for M draws from the predictive distribution at that location, we use the sample estimator of the kernel representation [Gneiting and Raftery, 2007],

$$\widehat{\text{CRPS}}(x_{1:M}; z) = \frac{1}{M} \sum_{i=1}^M |x_i - z| - \frac{1}{2M^2} \sum_{i=1}^M \sum_{j=1}^M |x_i - x_j|,$$

in which the first term rewards draws close to the observation and the second penalizes an overly dispersed predictive distribution. The same estimator, applied to each method’s own predictive draws, is used for every emulator, so no method is scored under a distributional assumption that another is not. All three depend on the predictive distribution only through its marginals, and so cannot detect whether a method reproduces dependence across locations.

To assess that dependence we use the variogram score of order p [Scheuerer and Hamill, 2015]. For a test input, let $z_1, \dots, z_L$ denote the held-out simulator output at the L spatiotemporal locations and let $\mathbf{X} = (X_1, \dots, X_L)$ be a random vector following the emulator’s predictive distribution at those locations. Then

$$\text{VS}_p = \sum_{i < j} (|z_i - z_j|^p - \mathbb{E}|X_i - X_j|^p)^2.$$

For each pair of spatiotemporal locations, the score thus compares the size of the difference between the two locations in the held-out simulator run, $|z_i - z_j|^p$, with the size predicted by the emulator, $\mathbb{E}|X_i - X_j|^p$. We take $p = 1/2$, which gave the best discrimination in the experiments of Scheuerer and Hamill [2015]. The expectation is estimated by Monte Carlo averaging over draws of $\mathbf{X}$, taken from each method’s own joint predictive distribution where one is available, and simulated independently across locations for methods that specify no cross-location dependence (e.g., MF-IDP and MF-FFN). The score is computed jointly over the full spatiotemporal domain for each test input, so a method that reproduces the marginals but not the cross-location dependence is penalized. The variogram score is complementary to AE; because it depends only on differences between locations, it cannot detect a bias shared by all locations [Scheuerer and Hamill, 2015], which AE does detect. Lower values are better for absolute error, CRPS, and the variogram score, while coverage is best near its 95% nominal level.

E.3 Rank Selection

The tensor-based methods selected mode-specific ranks to capture 95% of the explained variance. In this simulation study, this resulted in $(r_s, r_m, r_y, r_x) = (4, 4, 3, 11)$ for the LF-tensor and $(r'_s, r'_m, r'_y, r'_x) = (18, 8, 4, 8)$ for the discrepancy tensor. For MF-PCA, the number of components was chosen to explain the same percentage of variance as MF-TEN. The sampler settings and convergence diagnostics used for all MCMC-based methods are given in Section E.4.

E.4 MCMC Details

The following description highlights the key elements of the Markov chain Monte Carlo (MCMC) sampler used. Posterior samples for the parameters in model (8) were obtained using adaptive Metropolis–Hastings (MH) updates. The parameters updated include:

- λ_η : precision parameter for the approximation error $\mathcal{E}_\eta$, with an independent $\Gamma(1, 1/2)$ prior.
- λ_w : precision parameters $\lambda_1, \dots, \lambda_{r_x}$ for GPs placed on the weights, each with independent folded-normal priors with location 0 and scale 1.
- ρ : length scale parameters $\rho_{i\ell}$ for $i = 1, \dots, r_x$ and $\ell = 1, \dots, p$, each with independent log-normal priors with location 0 and scale 1.

The same hyperpriors were used for the precision and length scale parameters associated with the discrepancy weights in the MF setting.

At each MCMC iteration, parameters were updated sequentially using truncated normal proposal distributions: λ_η first, followed by each element of λ_w and ρ , with each proposal accepted or rejected via the MH acceptance ratio criterion before moving to the next parameter. In this work, the sampler was run for 10,000 iterations, discarding the first 5,000 as burn-in.

Additional Details

- During burn-in, proposal standard deviations were adapted every 50 iterations to target an acceptance rate of 0.45.
- Positive-definiteness of the GP covariance matrix was enforced using a near positive definite adjustment if necessary. Specifically, we used the `nearPD` function from the `Matrix` package in R. This function utilizes the algorithm developed by [Higham \[2002\]](#).
- Four chains were sampled, and convergence was assessed using both within-chain and between-chain diagnostics.
 - *Within-Chain*: Geweke Z-scores and p-values were computed by comparing the first 10% and last 50% of each chain. A Bonferroni correction was applied for multiple testing across chains, and adjusted p-values below $\alpha = 0.05$ were flagged as non-convergent.
 - *Between-Chain*: Gelman–Rubin diagnostics were conducted, ensuring that $\hat{R} < 1.1$ for all parameters.
- For the multi-fidelity hierarchical model outlined in (11), a similar process was followed with additional updates for λ_δ and the GP hyperparameters (precision and length scale) for the discrepancy components.

E.5 Input Sensitivity

Table S1 reports the posterior mean length scale for each input in the MF-TEN emulator, averaged over the low-fidelity weights and over the discrepancy weights respectively. Length scales indicate how sensitive the model output is to changes in each input, with smaller values corresponding to higher sensitivity. As expected, the dummy input x_3 had no effect on either the LF or discrepancy weights, and the emulator correctly assigned it a larger length scale than the two active inputs: for the low-fidelity weights it is roughly three times that of x_1 and x_2 , which are themselves nearly identical. We note that this separation is weaker for the discrepancy, whose output depends only

Table S1: Posterior mean length scales for each input in the MF-TEN emulator from the simulation study, averaged over the low-fidelity weights w_i and over the discrepancy weights v_i respectively. Smaller values indicate greater sensitivity to that input.

	Low-Fidelity	Discrepancy
x_1	0.49	1.92
x_2	0.49	1.23
x_3	1.52	2.10

on the spatiotemporal values and not on the input parameters, so that no input is strongly active and the length scales for x_1 and x_3 are comparable.

F Basis Uncertainty

In our manuscript, we treat the spatiotemporal bases $\mathcal{B}$ as fixed quantities estimated from the data. We have not implemented a method that fully accounts for the uncertainty in their estimation. A natural extension would be to place a prior on $\mathcal{B}$ and perform joint inference over $\mathcal{B}$ and the remaining parameters, enabling uncertainty propagation to posterior estimates and predictions [Zhao et al., 2015, Stolf and Canale, 2025]; this would, however, introduce additional computational complexity due to the need to sample from a higher-dimensional posterior involving the basis parameters. What we do instead, and what this section reports, is an empirical assessment of how stable the estimated bases are.

For the MPAS-Seaice application, we assess how stable the estimated bases are under resampling of the design ensemble. We draw 1,000 bootstrap resamples of the design points, re-estimate the Tucker decomposition on each, and for every mode compare the resampled subspace to the full-data subspace through the Frobenius distance between their orthogonal projection matrices,

$$d(U^{(b)}, U^{(0)}) = \|U^{(b)}U^{(b)\top} - U^{(0)}U^{(0)\top}\|_F,$$

where $U^{(0)}$ is the factor matrix estimated from the full data and $U^{(b)}$ that from the b -th resample. This distance depends only on the spans of $U^{(b)}$ and $U^{(0)}$ and is therefore invariant to the sign, ordering, and rotational ambiguities of the factor matrices. Because each factor matrix has orthonormal columns, $UU^\top$ is the orthogonal projection onto a subspace whose dimension r is the number of columns of U , and

$$d(U^{(b)}, U^{(0)})^2 = 2 \sum_{i=1}^r \sin^2 \theta_i,$$

where $\theta_1, \dots, \theta_r$ are the principal angles between the two subspaces. The distance is therefore bounded below by 0, attained when the subspaces coincide, and above by $\sqrt{2r}$, attained when they are orthogonal. For the MPAS-Seaice decomposition the relevant ranks are $r_s = 79$, $r_m = 7$, and $r_y = 6$, giving upper bounds of 12.6, 3.7, and 3.5 for the three modes. Table S2 reports the mean and standard deviation of $d(U^{(b)}, U^{(0)})$ across resamples for the spatial, monthly, and yearly modes. The distances are several orders of magnitude below these bounds for all three modes, corresponding to $\sum_i \sin^2 \theta_i < 10^{-4}$ in every case. This indicates that the estimated subspaces, and hence the plug-in bases $\mathcal{B}$, vary little under resampling. We emphasize that this is a stability assessment and not a full propagation of basis-estimation uncertainty into the posterior predictive distribution. The reported credible intervals are therefore conditional on the estimated bases, and should be interpreted as such.

Table S2: Bootstrap assessment of mode-subspace stability. For each mode we draw bootstrap resamples of the design points, re-estimate the Tucker factor matrices, and compute the projection distance $d(U^{(b)}, U^{(0)}) = \|U^{(b)}U^{(b)\top} - U^{(0)}U^{(0)\top}\|_F$ between each resampled subspace and the full-data subspace. The table reports the mean and standard deviation of these distances across resamples; small values indicate stable mode subspaces.

Mode	Mean distance	SD of distances
Space	0.006	0.002
Month	0.011	0.007
Year	0.011	0.005

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